



Introduction to Psychology- I

LA380 / LAL320

Topics to be Covered

- Biology of Behaviour: Species typical behaviour patterns – Brain and behaviour – neurons and synapses – the nervous system and its functions in behaviour- hemispheric functions- neural degeneration and neural regeneration- blocking neural degeneration
- Principles of learning – definition and perspectives of learning, classical conditioning, instrumental conditioning- Thorndike, Skinner- reinforcement-schedules of reinforcement- role of aversive stimuli- applications- cognitive learning theory - learning and the brain

Species typical behaviour patterns

- Ethology and Ethologists: Study of behaviour patterns typical of particular species of animals- with emphasis on the evolution of these patterns and adaptive value.
- Species Typical behaviour:
 - All normal members must display the behaviour under certain circumstances.
 - Arise from genetic heritage of the species evolved over time – part of “species nature” can serve social, reproductive, nutritional, or other functions within the context of an animal’s natural history (Huskisson, 2019).
 - Environment plays a role in developing and perfecting it –
 - These behaviours are relatively fixed and inflexible patterns of behaviour (movement) trigger by a particular stimulus or event in the environment- storing food in the winters by desert rodents, hibernation in winters



Species typical behaviour patterns

- Releaser
- Fixed action pattern : behaviour released by such a stimulus
- E.g. Stickleback fish – given the stimulus, the behaviour will follow
- May seem limiting from the human perspective, but for the animals it is adaptive



Species typical behaviour among cowbirds



A cowbird

Promotes finding mates in own species to improve chances of species survival

Survival can be promoted by warning/threatening

- **Species typical behaviour can also be in form of aggression** – dogs- curling lips and barring teeth to show threat



Emotional expression is mostly same in all members of a particular species

- Crucial for social interactions of species members
- Among humans it provides signals about other animal's or person's emotional state and consequently behaviour that may be forthcoming

Species typical behaviour among Humans

- More complex than lower animals – due to the flexibility of behaviour
- Human evolution – lead to – behaviour strongly influenced by learning – through events that take place in life
- **Fixed action behaviours** – prominent in lower animals not so prominent in human nature
- Infants reflexes- grope, swallow, breathing, blinking, coughing, sneezing, and withdrawing from painful stimuli
- Moro reflex: when support is suddenly removed from behind the head, the neonate flings arms to the side, extend its fingers and bring arms inward in a sort of embracing movement



Expression of certain emotions are species typical behaviours in humans



Happiness



sadness



surprise



fear



anger



disgust

While not prominent, we do have some species specific behaviours as part of our animal natures.

Evolutionary heritage leading to behaviours in Humans

- Evolution effects at a deeper and fundamental level – predisposes us to act or exhibit certain behaviours as opposed to others.
 - More probable or less probable
 - **Built-in programmed biases towards certain types of behaviour**– e.g altruism – the lookout calling attention to themselves so that others can run away
 - more genes of altruism are ultimately persevered than removed from the death of one single altruistic individual – making groups more adapted to survival- thus altruistic gene gets selected and becomes part of the species gene pool/ heritage.
- Certain behaviours are learned – produce and understand language – no matter which language we speak
 - These are broad tendencies – not specific behaviours (which are based on individual experiences- choosing to follow a career after exposure to adverse conditions)

Books

- Atkinson., Hilgard., Nolen-Hoeksema, S., Fredrickson, B., Loftus, G. and Lutz, C. (2014). Introduction to psychology. Hampshire, United Kingdom: Cengage Learning EMEA.
- Morgan, Clifford T. & King, Richard A. (2017). Introduction to psychology. New York: McGraw-Hill
- Saundra K. Ciccarelli, Girishwar Misra, J. Noland White. Psychology (5e). Pearson Education India

Tendencies towards competition and aggression- especially in males is innate

Territoriality, need for organization with leaders, certain child rearing behaviours, male female roles – a thought to be innate behavioural potentialities- having a basis in evolution

This is controversial- though anthropological evidence of stone age cultures support innate basis of behavioural tendencies- e.g. aggression- other evidences do not.

Evidence for nurture- elder Eskimos choosing to stay behind and die rather than slowing the whole group, search and rescue parties looking for lost persons even under dire circumstances.

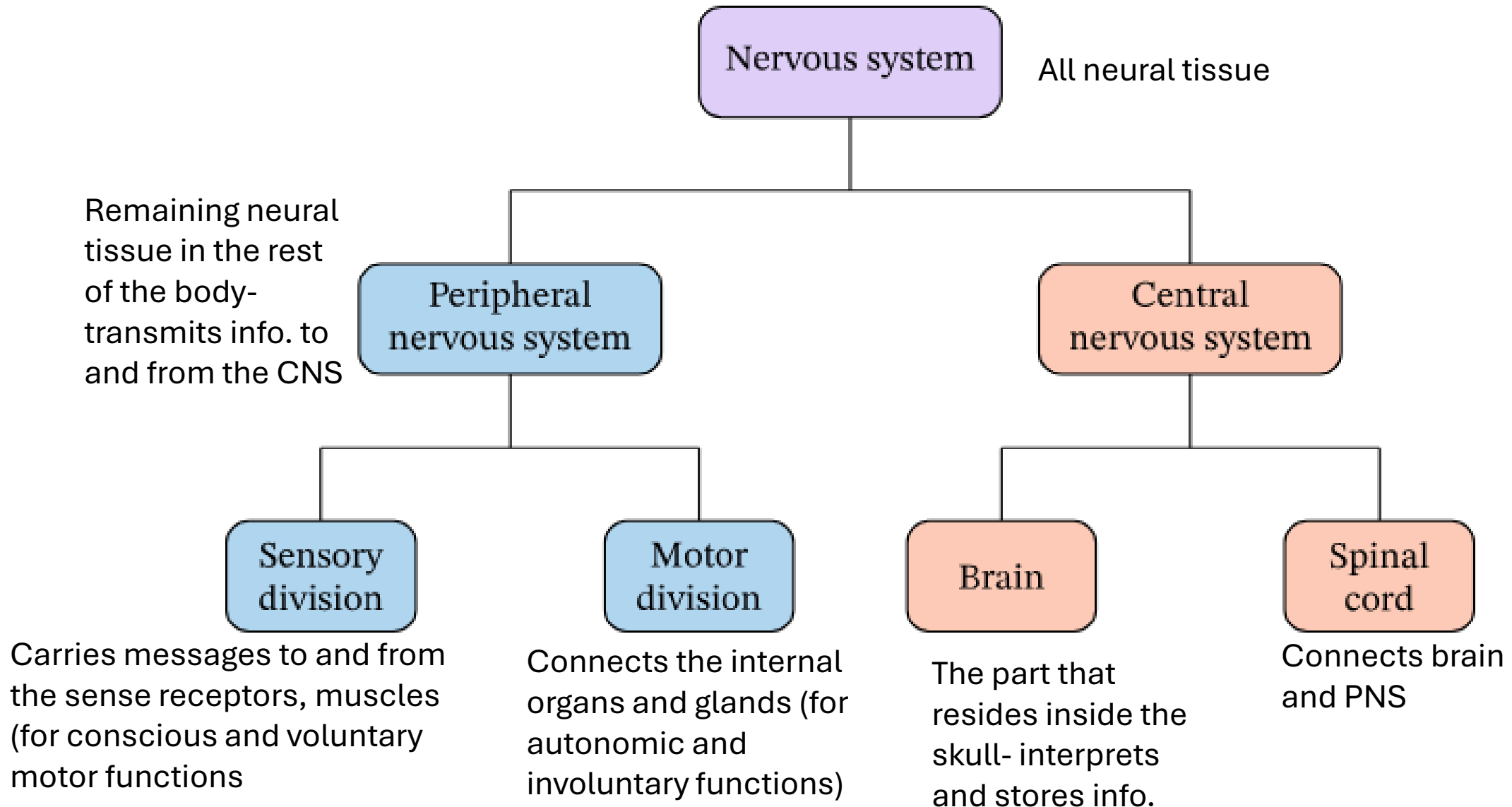
Nature vs. Nurture

- In the absence of conclusive evidence the debate about innate vs environmental influences on human behaviour continues.
- Psychologists emphasise both – genes and structures as well as unique life circumstances that undermine human nature
- One way to look at biological and evolutionary basis of behaviour- Ernst Mayr (1960) – many evolutionary “novelties” result due to co-opting existing systems for new functions
- **The voicing system-** teeth, tongue, mouth initial purpose – eating – evolved to produce sounds and later the pronunciation of language (Atkinson & Hilgard, 2014)
- Similarly, the disgust response described by Darwin- the gape, tongue extension, nose wrinkle and dropping of the mouth corners developed as response to preventing food from going inside the mouth or encourage its discharge. **Disgust response is associate with the brain’s right prefrontal area and basal ganglia.** Rozdin et al. (2000) propose what was to avoid harm to the body evolved to avoid harm to the soul- against morally disgusting. But learning what is morally disgusting is part of socialization in cultures.
- Look at the structures of the brain and the nervous system:
CASE



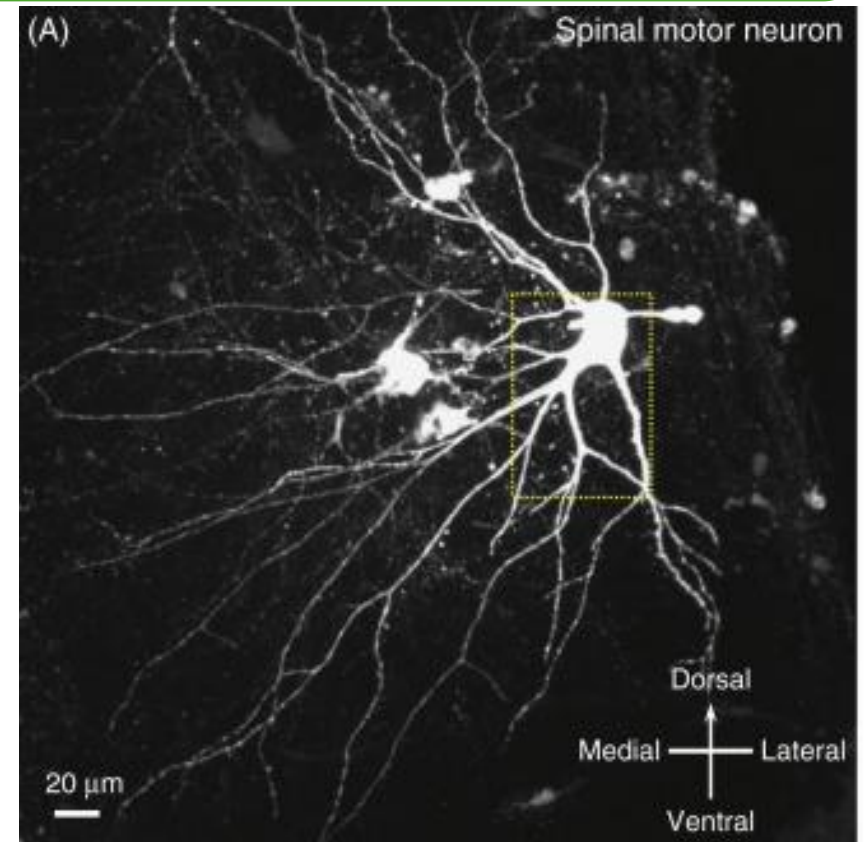
Facial expression of disgust

Organization of the Nervous System



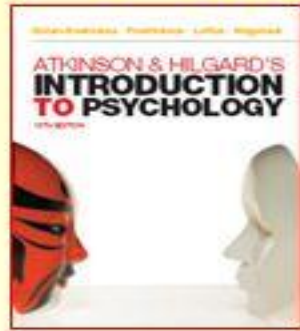
In the Somatic System/ Sensory division

- Sensory nerves:
 - Transmit information about the external stimulation from the skin, muscles, and joints to the CNS
 - We become aware of pain, pressure, temperature variations
- Motor nerves:
 - Carry impulses from the CNS to the muscles, where they initiate action
 - Control all muscles we use in voluntary movements as well as involuntary adjustments in posture and balance



Mentis (2013)

Neurons, the Building Blocks of the Nervous System



- Basic unit of the nervous system = Neuron
 - Specialized cell that transmits neural impulses or messages to other neurons, glands and muscles

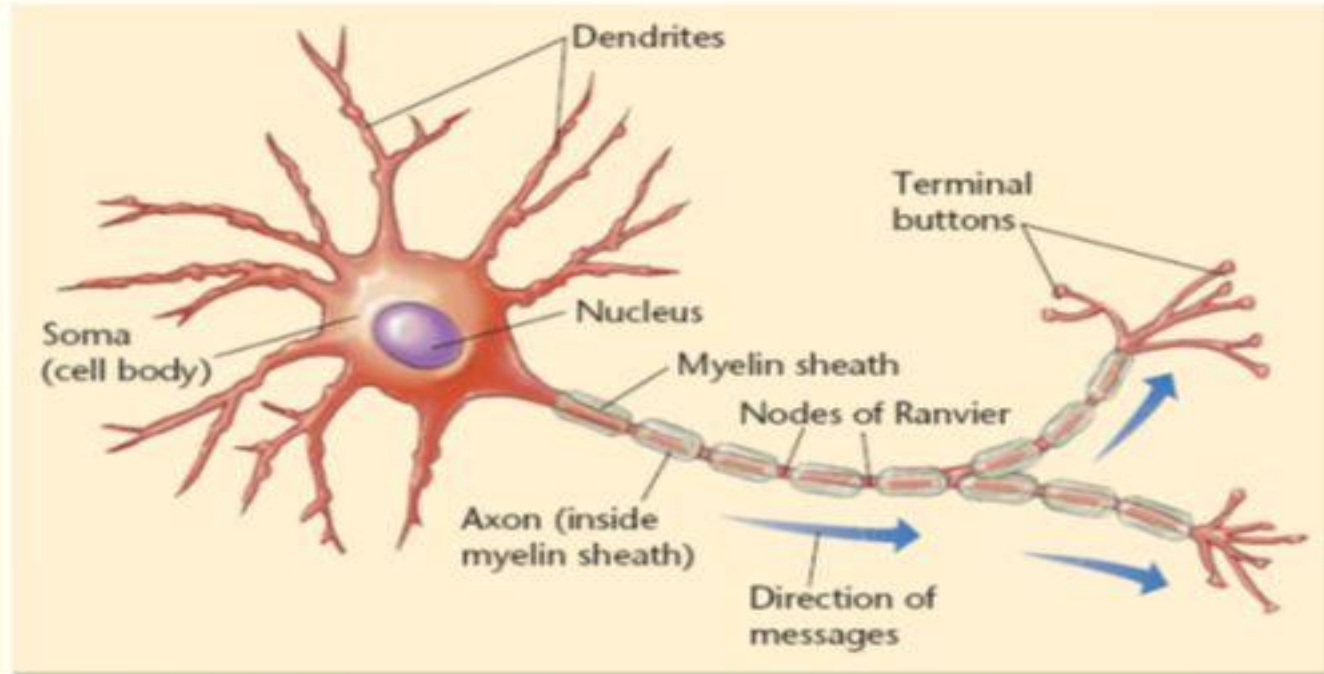
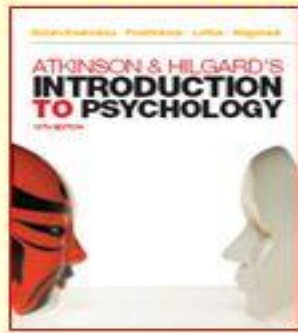


Figure 2.2 Schematic Diagram of a Neuron.

Arrows indicate the direction of the nerve impulse. Some axons are branched; the branches are called collaterals. The axons of many neurons are covered with an insulating myelin sheath that helps increase the speed of the nerve impulse.

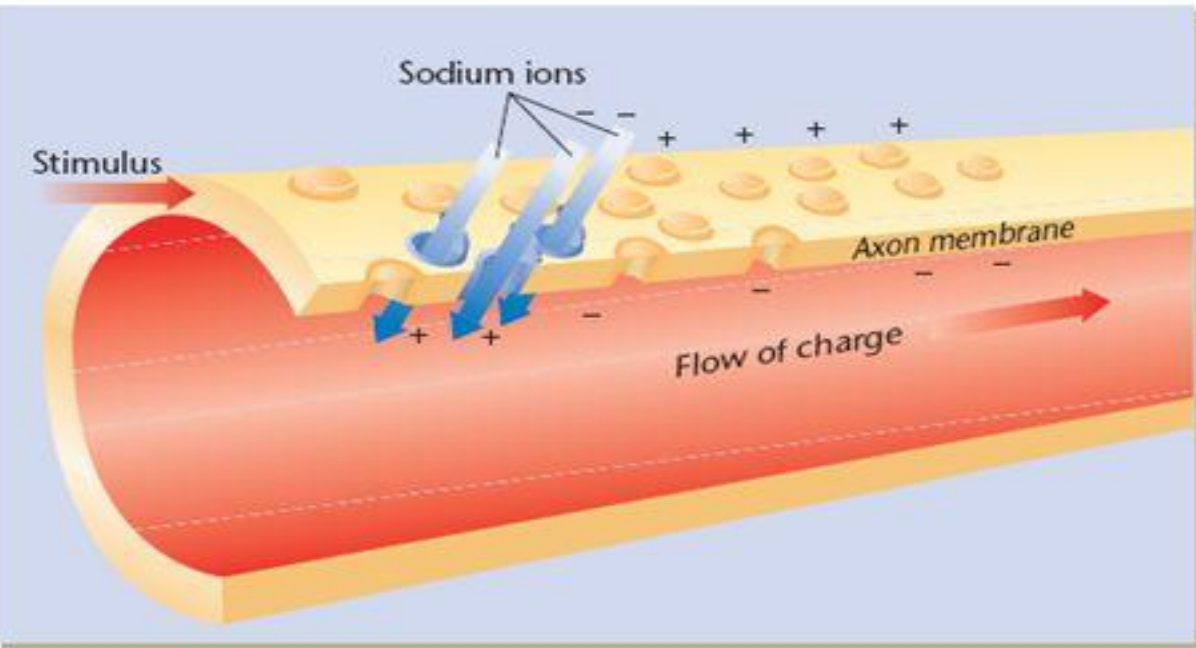
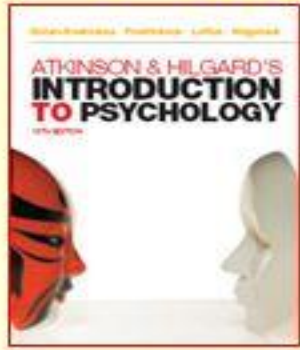
(Adapted from Human Anatomy by Anthony J. Gaudin and Kenneth C. Jones. Copyright © 1988 by Anthony J. Gaudin and Kenneth C. Jones. Reprinted by permission of the authors.)

Neurons, the Building Blocks of the Nervous System



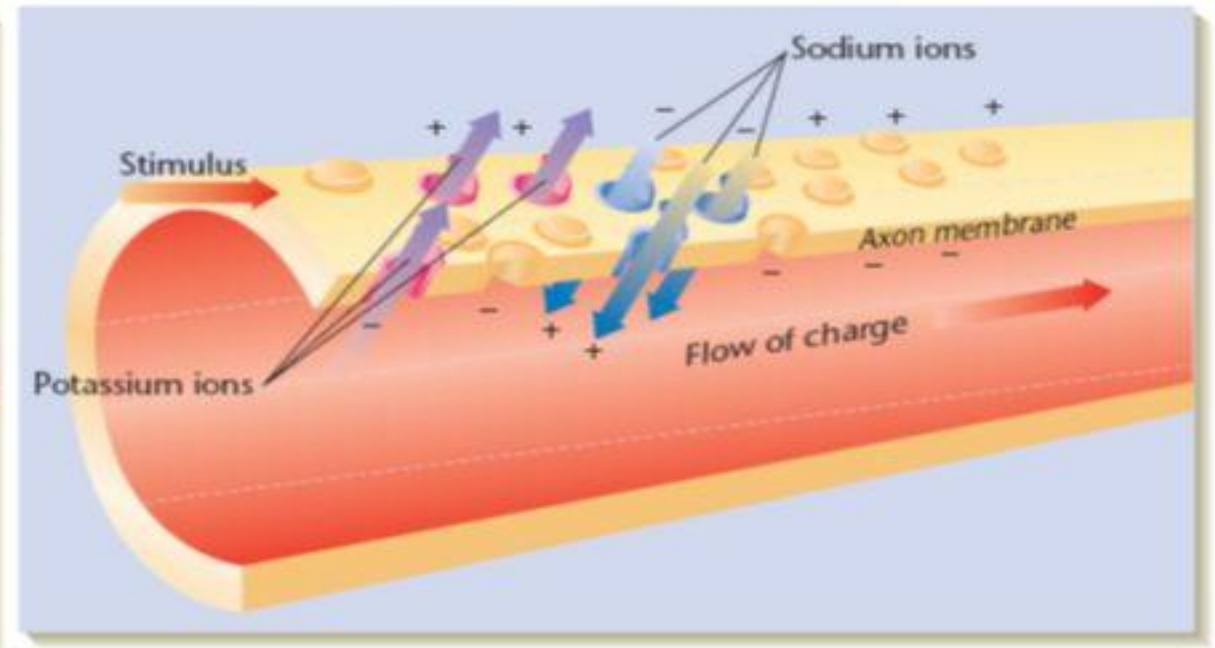
- Action Potentials
 - Information travels along neuron in form of neural impulse called an action potential
 - Each action potential is the result of movement of electrically charged particles (ions) in and out of neuron
 - Movement of ions controlled by ion channels - when these are closed the neuron is referred to as “resting”
 - When a neuron is depolarized above the excitation threshold cell membrane is temporarily unstable, resulting in action potential

Neurons, the Building Blocks of the Nervous System



a) During an action potential, sodium gates in the neuron membrane open and sodium ions enter the axon, bringing a positive charge with them.

Resting potential: -70 millivolts
Excitation threshold: -55 mV

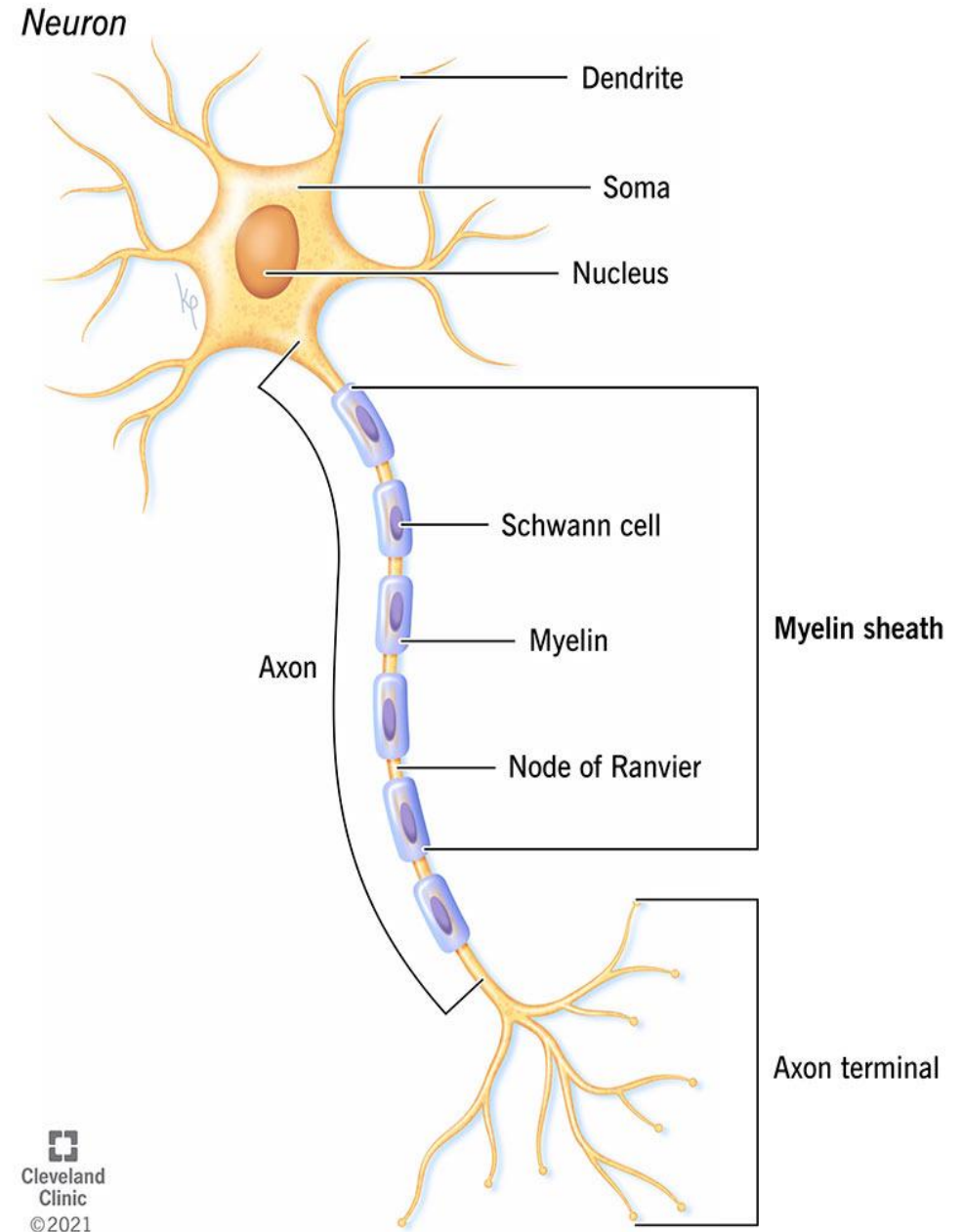


b) After an action potential occurs at one point along the axon, the sodium gates close at that point and open at the next point along the axon. When the sodium gates close, potassium gates open and potassium ions flow out of the axon, carrying a positive charge with them.

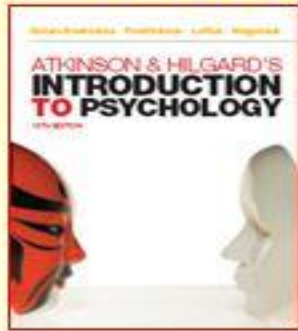
Inside goes up to +40 mV

Figure 2.7 Action Potential Propagating along the Axon.

- Why does the action potential travel in only one direction?- due to a **refractory period**- after it had generated a “spike” the membrane cannot do so again for about a millisecond.
- Speed: 1- 120 meters/ second – affected by cover of myelin sheath
- The myelin sheath is particularly prevalent where rapid transmission of the action potential is critical- e.g. – long axons that stimulate skeletal muscles
- In multiple sclerosis , a disorder in which symptoms first become evident between the ages of 16-30, the immune system attacks and destroys body’s own myelin sheaths, - resulting – in severe motor nerve dysfunction



Neurons, the Building Blocks of the Nervous System



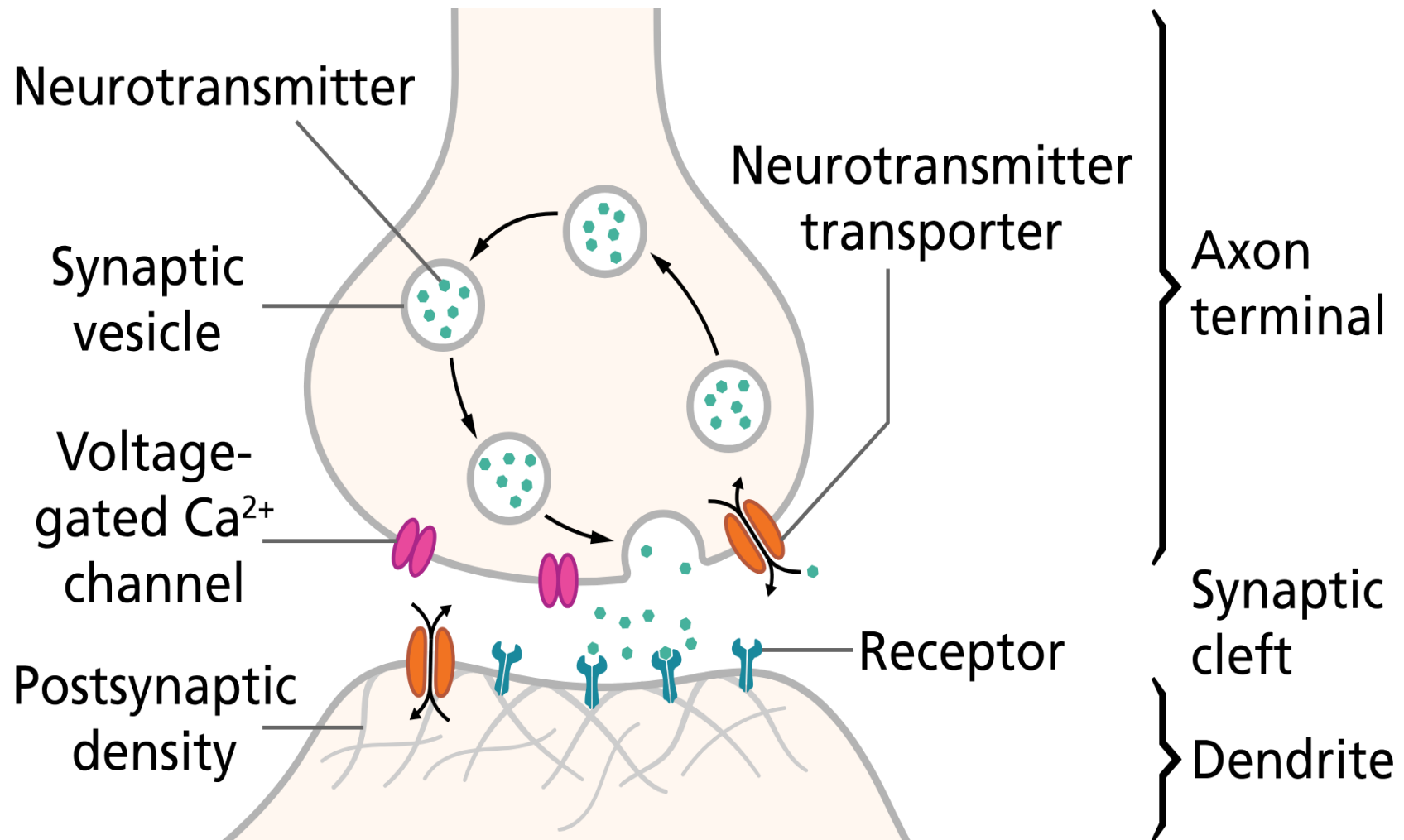
- **Synaptic** transmission and neural coding
 - All-or-none law – a neuron is either firing action potential, or not, so how can the nervous system code for/represent complexity of experience?
 - Power of nervous system lies in the complexity of the *connections* between neurons
 - Lock-and-key action – the way a neurotransmitter and receptor fit together, which causes a change in permeability of ion channels, either excitatory or inhibitory

- Another way- involving a greater population of neurons in the response
- Population coding can be powerful- synchronization (or lack of it) in response to the individual neurons often contains meaning as well.
- What becomes important therefore is the connections between the individual neurons.

Synaptic Transmission

Action must be brief to maintain precise control

1. Reuptake
2. Degradation



2 types of effects

- **Excitatory effect:** allows positively charged ions (such as Na^+) to enter the post synaptic neuron – resulting in depolarization of the neuron.
-This makes the post synaptic neuron more likely to reach its excitatory threshold and thus more likely to generate an action potential.
- **Inhibitory effect:** positively charged ions (K^+) leave the neuron or a negatively charged ions (Cl^-) enter – making the inside more negatively charged than before.
- Consequently less likely to reach the excitatory threshold and therefore less likely to generate an action potential

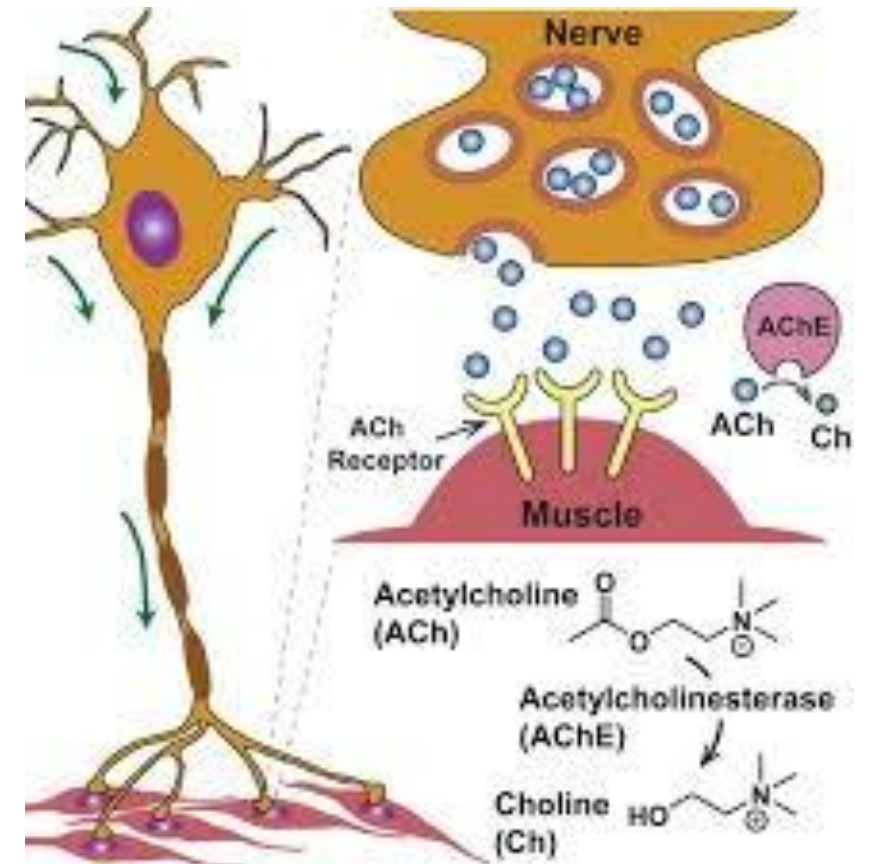
Neurotransmitters : Acetylcholine

- Present at many synapses throughout the nervous system
- Usually excitatory, sometime inhibitory – depending on the type of receptor molecule in the membrane of the receiving neuron.
- High concentration: Forebrain- hippocampus – plays a key role in the formation of new memories (Eichenbaum, 2000).
- Related to Alzheimer's disease- a memory disorder. – neurons in the forebrain that produce acetylcholine degenerate in Alzheimer's patients.
- The less acetylcholine is produced the more is the memory loss.



Acetylcholine

- Released at every synapse between a motor neuron and a skeletal muscle fibre- causing muscle contraction
- Drugs can affect acetylcholine and produce muscle paralysis
 - Botulinum toxin: bacteria from improperly canned foods blocks release of acetylcholine and causes death by paralysis then muscles used in breathing
 - Nerve gases and many pesticides: cause paralysis by destroying the enzyme that degrade acetylcholine- causing uncontrolled buildup and normal synaptic transmission is impossible

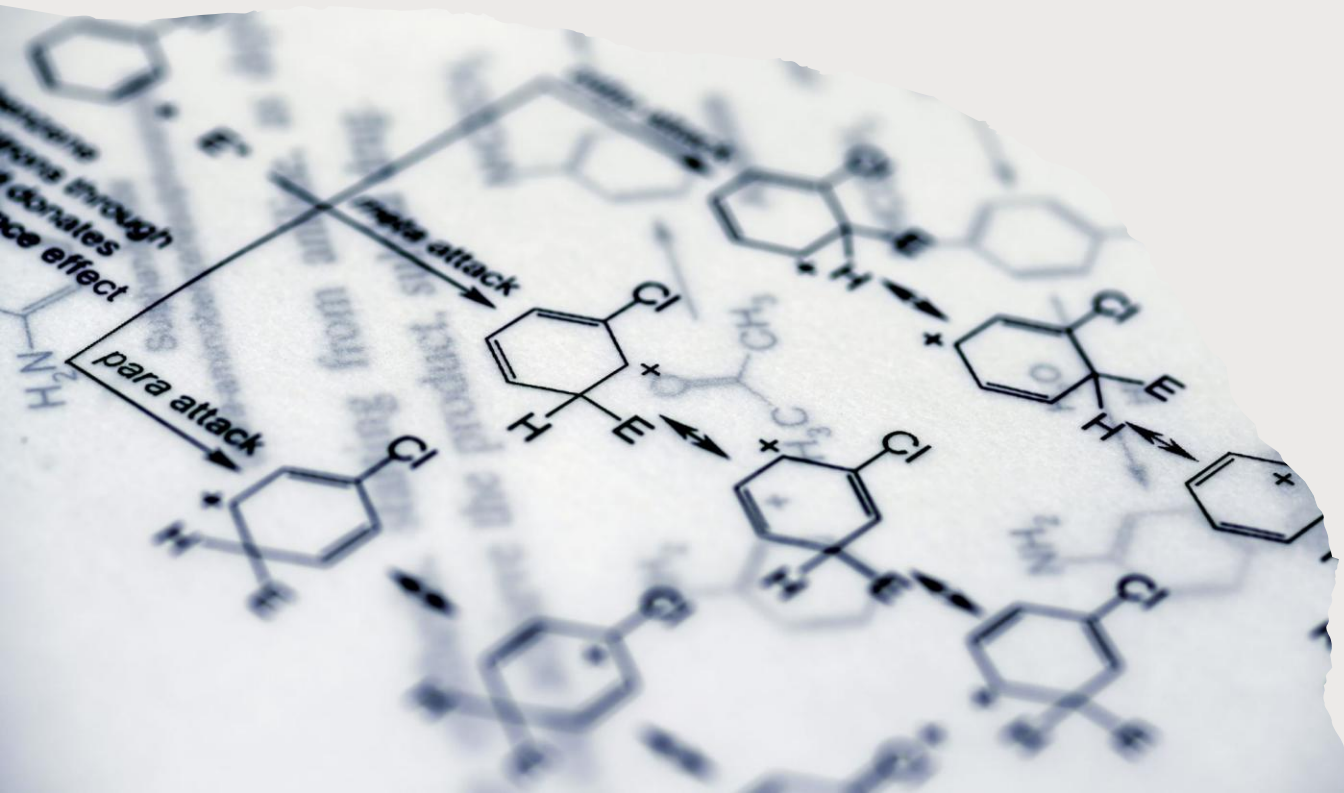


Norepinephrine

- Noradrenaline
- Produced by neuron of the brain stem
- **Alertness and arousal**
- Any drug that causes an increase or decrease in this neurotransmitter is correlated with increase or decrease in the individual's mood level
- Cocaine and amphetamines prolong the action norepinephrine by slowing its reuptake – receiving neurons activated for a longer time
- Lithium speeds up reuptake – causing mood to be depressed

Dopamine

- Chemically similar to norepinephrine
- Regulates motivation and emotional pleasure (e.g. food and sex; goal attainment)
- Regulates incentive salience (Schultz, 2002) – how much we want something : e.g. food (Volko et al., 2012)
- Too much dopamine in some parts of the brain may cause schizophrenia (drug: clozapine or chlorpromazine)
- Too little in other parts causes Parkinson's disease (L-dopa)





Serotonin

Mood regulation

Low levels: feeling depression

Serotonin reuptake inhibitors are used as anti depressants that increase serotonin in the brain by blocking its uptake.
(Prozac, Zoloft, Paxil)

Important in regulation of sleep and appetite – used to treat eating disorder of bulimia

Lysergic acid diethylamide (LSD) induces the effects of serotonin by binding to its receptors in the brain

GABA

Gamma-aminobutyric acid

Major inhibitory transmitter

Most synapses in the brain use GABA

Blocking of GABA receptors produces convulsions as muscle movements cannot be controlled without GABA's inhibitory influence.

Tranquilizing effects of certain anti-anxiety drugs (benzodiazepines) are a result of GABA's inhibitory actions

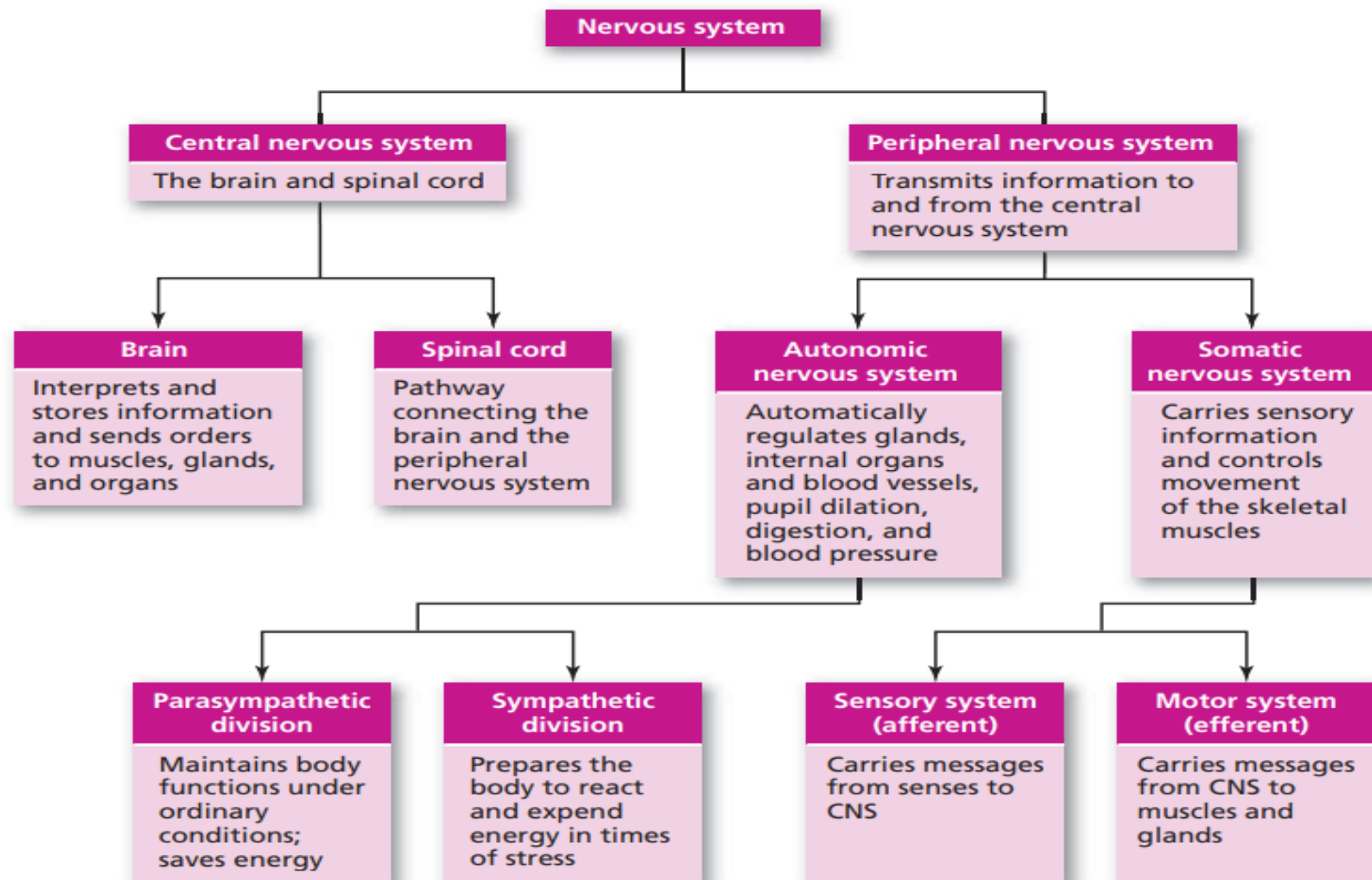
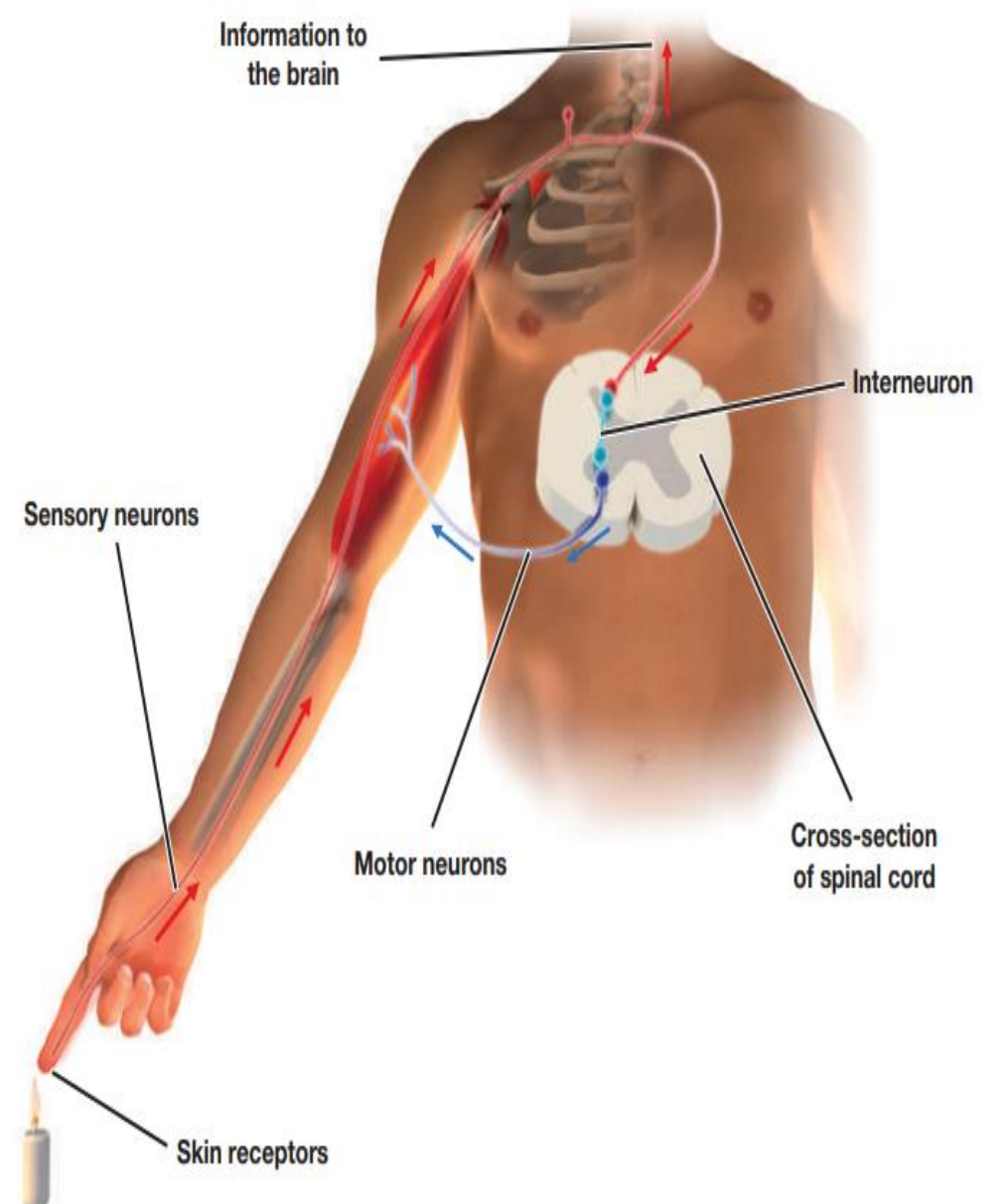


Figure 2.5 An Overview of the Nervous System

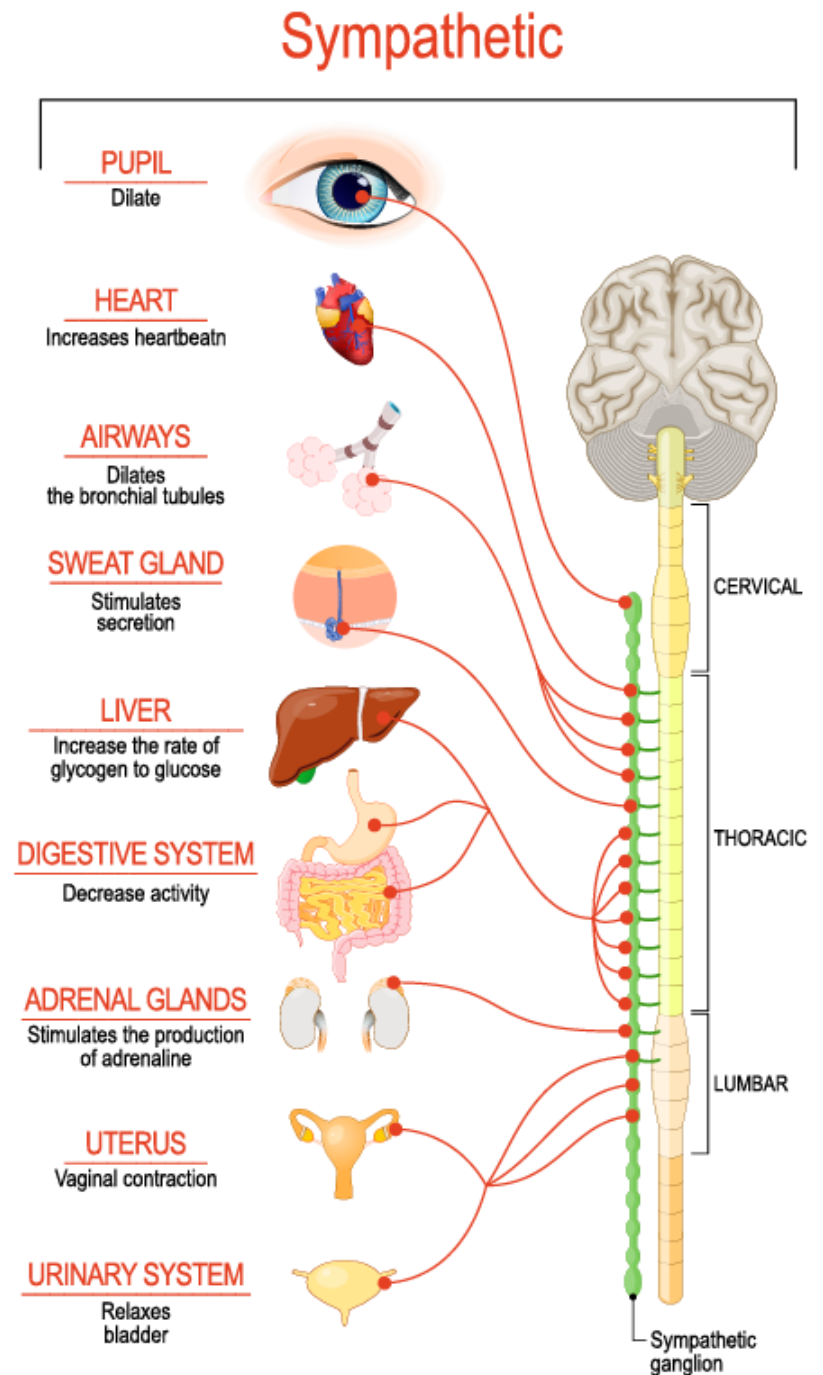
The Spinal Cord

- Long bundle of neurons that serves two vital functions.
 - Outer section is composed mainly of myelinated axons and nerves, which appear white, whereas the inner section is mainly composed of cell bodies of neurons, which appear gray.
1. The purpose of the outer section is to carry messages from the body up to the brain and from the brain down to the body. It is simply a message “pipeline.”
 2. **Reflexive action: Fast response time – Reflex arc**
 - The inside section, which is made up of cell bodies separated by glial cells, is actually a primitive sort of “brain.”
 - This part of the spinal cord is responsible for certain reflexes—very fast, lifesaving reflexes.
 - Touch a flame or a hot stove with your finger, for example, and an afferent neuron will send the pain message up to the spinal column, where it enters into the central area of the spinal cord. The interneuron in that central area will then receive the message and send out a response along an efferent neuron, causing your finger to pull back. This all happens very quickly.



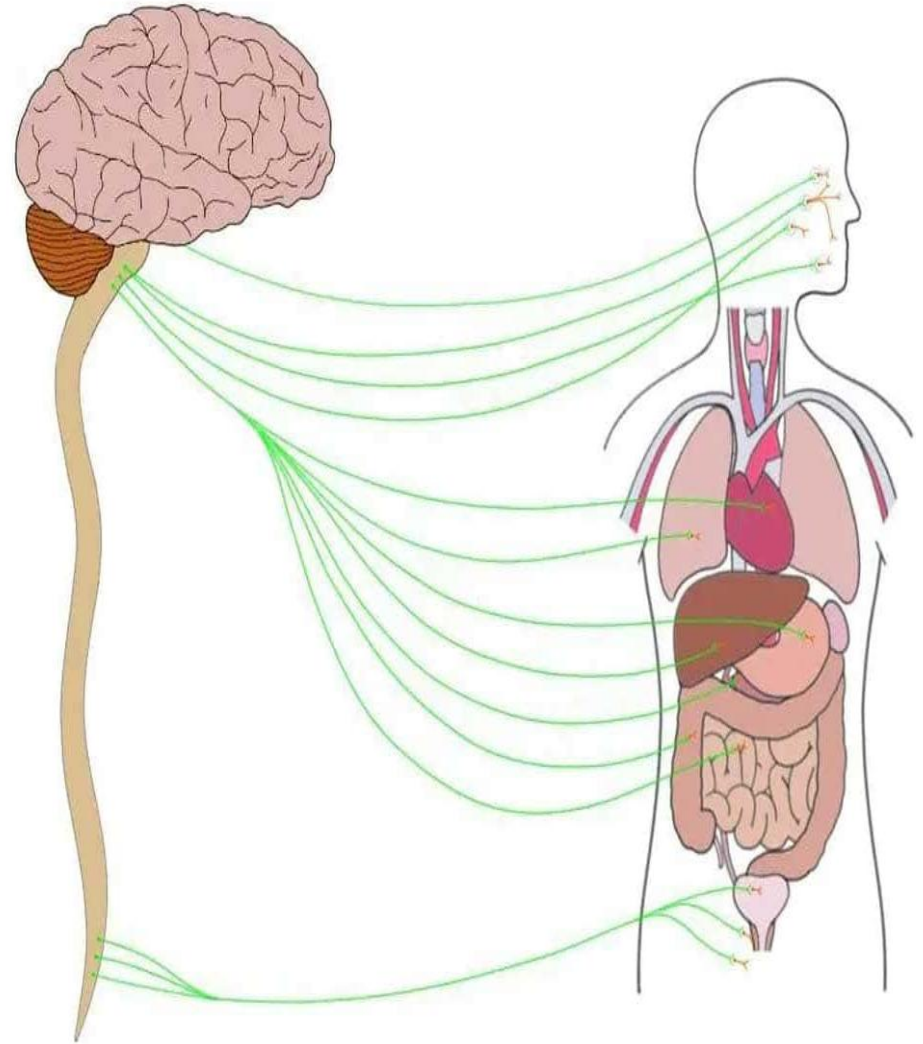
The Sympathetic Division

- Located on the middle of the spinal column —running from near the top of the ribcage to the waist area.
- Usually called the “fight-or-flight system” (sympathy with one’s emotions?)
- Get the body ready to deal with stress- anger or joy , fear
- ✓ Pupils get bigger
- ✓ Heart starts pumping faster, drawing blood away from non-essential organs (skin) sometimes brains (fainting)
- ✓ Lungs work faster, increase blood oxygen (Heavy breathing)
- ✓ Adrenal glands: release stress related hormones (adrenaline, cortisol – heart, lungs, and muscles)
- ✓ Not every organ stimulated- digestion, excretion, saliva dries up – really scared the bowels or bladder empty
- ✓ Burn blood sugar
- Stress ends- activation of parasympathetic division.
- Stress for too long- person collapses – as PNS division overreacts in trying to inhibit SNS- heart slows, blood vessels open up, blood pressure in the brain drops- resulting in fainting.



The Parasympathetic Division

- “eat-drink-and-rest” system
- Para = beyond or next; refers to neurons located on either side of the sympathetic division neurons
- Location; top and bottom of the spinal cord
- Two different functions:
 1. Brings body down to rest or normal functioning after a fight and flight response –
 - ✓ Slows heart rate and breathing
 - ✓ Constricts pupils
 - ✓ Reactivated digestion and excretion
 - ✓ Signals to the adrenal glands to stop (PNS not connected to adrenal glands)
 2. Responsible for most ordinary everyday functioning- regular heartbeat, normal breathing, and digestion
- At any given time- SNS or PNS determine if individuals are aroused or relaxed



REGENERATION AND REPAIR OF NERVOUS TISSUE

- Earlier: neurons don't repair
- E.g. Recovery from stroke- due to healthy brain cells taking over the functions of damaged cells.
- Throughout your life, your nervous system exhibits **plasticity**, the capability to change based on experience.
- Brain: **neuroplasticity- ability to constantly change both the structure as well as function of many cells in the brain in response to trauma or experience (Neville & Bavelier, 2000)**
- At the biological level: At the level of individual neurons, the changes that can occur include the sprouting of new dendrites, synthesis of new proteins, and changes in synaptic contacts with other neurons (Sanes & Jessel, 2013).
- **Neurogenesis: Brain can change through this process. Formation of new neurons and important process during the development of our nervous system.**
 - Mostly prior to birth – during prenatal period
- Regeneration : still limited in mammals - mostly hippocampus
- Physical exercise benefits neurogenesis and neuroplasticity (Prakash, 2015)
- Good sleep is important (Tesler, 2016)
- Stem cell transplantation – research ongoing (Huang et al, 2015)

Case: Phineas Gage

Principles of Learning

Learning: Learning may be defined as a relatively permanent change in behaviour that occurs as the result of prior experience. (Atkinson & Hilgard, 2019)

“Relatively permanent”: when people learn anything, some part of the brain is changed to record this (Farmer et al. 2013)

Process of memory- without the ability to remember what happens learning cannot take place.

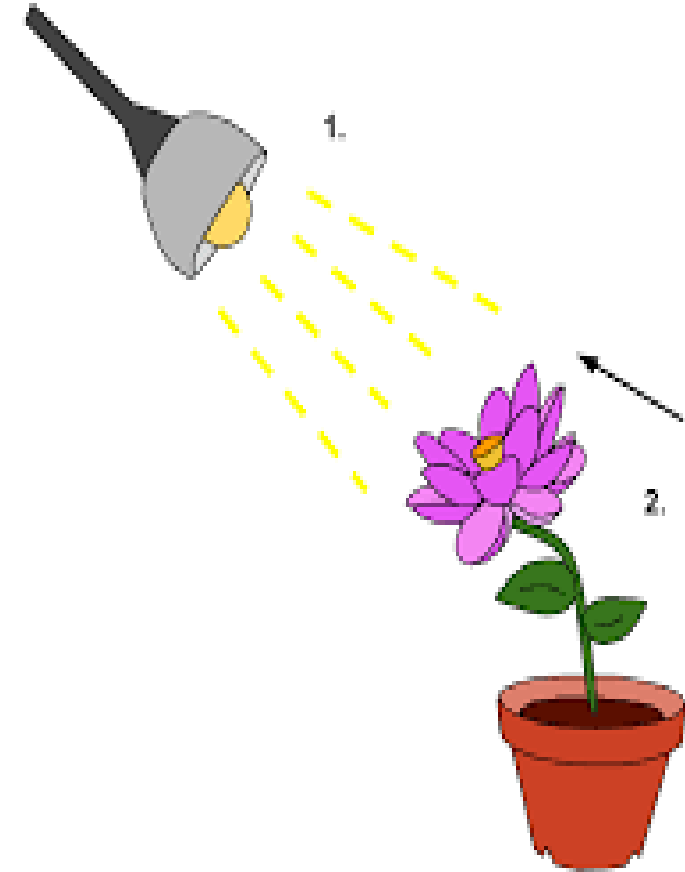
“experience or practice”: e.g. last time you experienced pain- we learn. Vice versa for pleasure experience.

Not all change is accomplished through learning

Maturation: height, size of the brain, practice alone will not allow a child to walk. – genetic blueprint

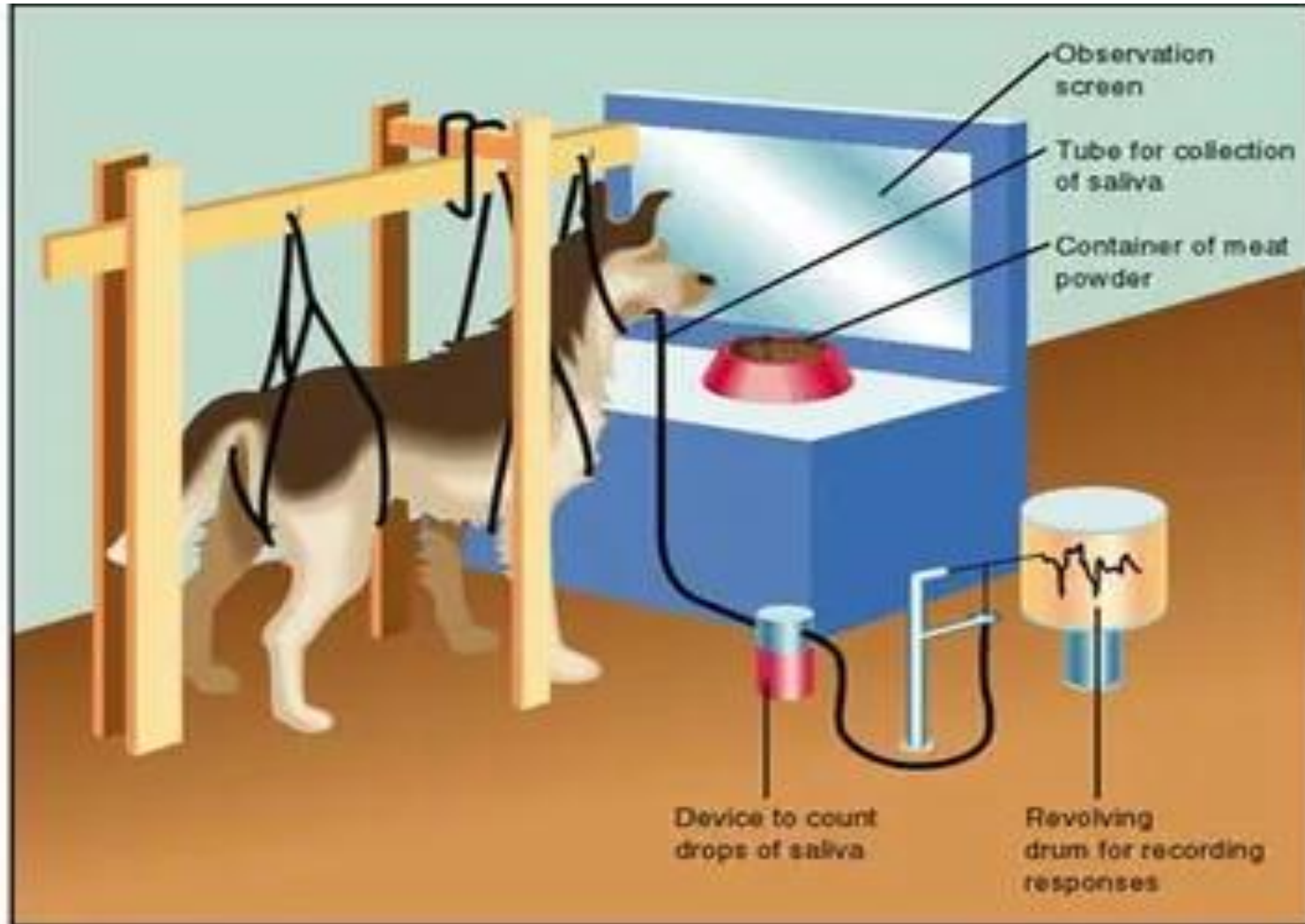
Classical conditioning

- A learning process in which a previously neutral stimulus becomes associated with another stimulus through repeated pairing with that stimulus.
- Salivary glands automatically start releasing saliva to help with chewing and digestion- normal **reflex**—an unlearned, involuntary response that is not under personal control or choice—one of many that occur in both animals and humans.
- The food causes a particular reaction, the salivation.
- A *stimulus* can be defined as any object, event, or experience that causes a *response*, the reaction of an organism.
- In the case of Pavlov's dogs, the food is the stimulus and salivation is the response.



Pavlov's Classical Conditioning/ Respondent Conditioning

Apparatus used by Ivan Pavlov



- **Unconditioned stimulus (= unlearned) :** (UCS) stimulus consistently evoke a response, often involuntary in nature, or is reliably followed by one, – e.g. food
- **Conditioned Stimulus:** (CS) – neutral initially- has only a function of alerting, or attentional the first few times – e.g. bell
- **Unconditioned response:** response following the UCS

UCR



CR

Before conditioning

**FOOD
(UCS)**

**SALIVATION
(UCR)**



BELL

NO RESPONSE



During conditioning

**BELL +
FOOD
(UCS)**

**SALIVATION
(UCR)**



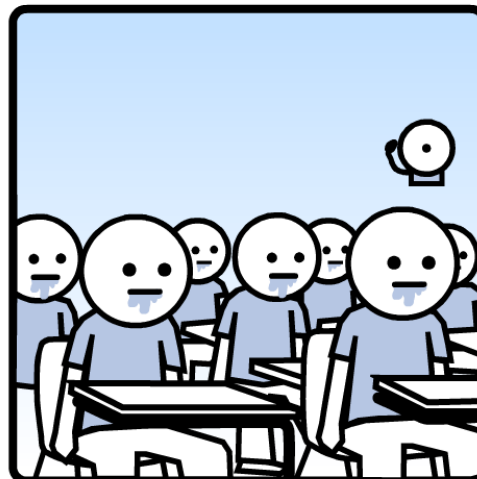
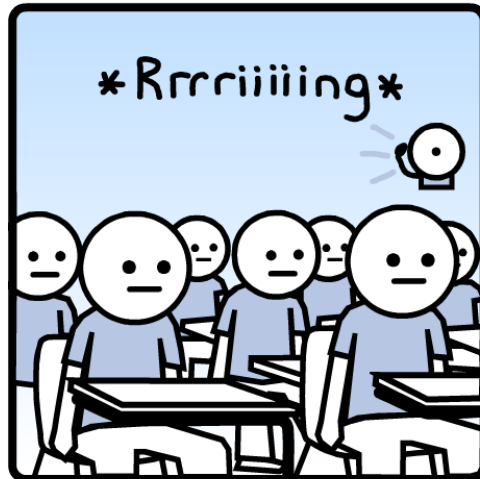
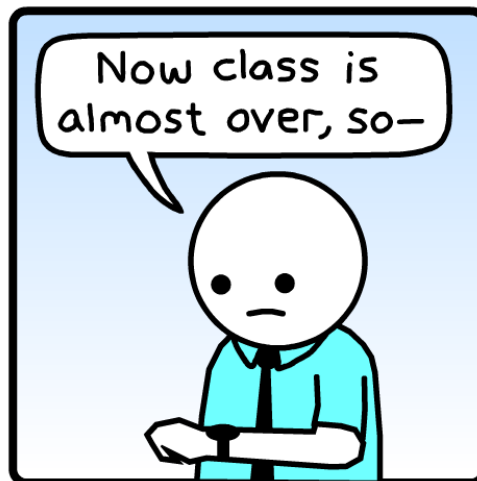
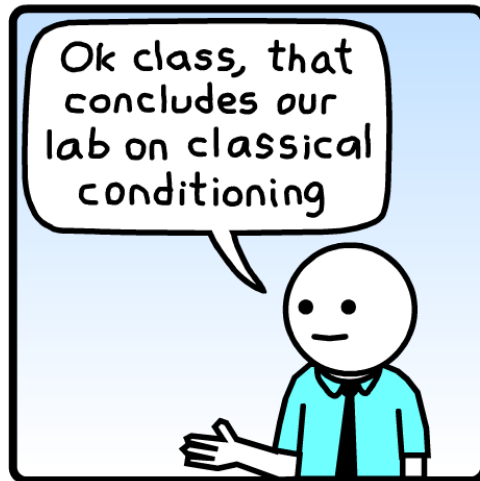
After conditioning

**BELL
(CS)**

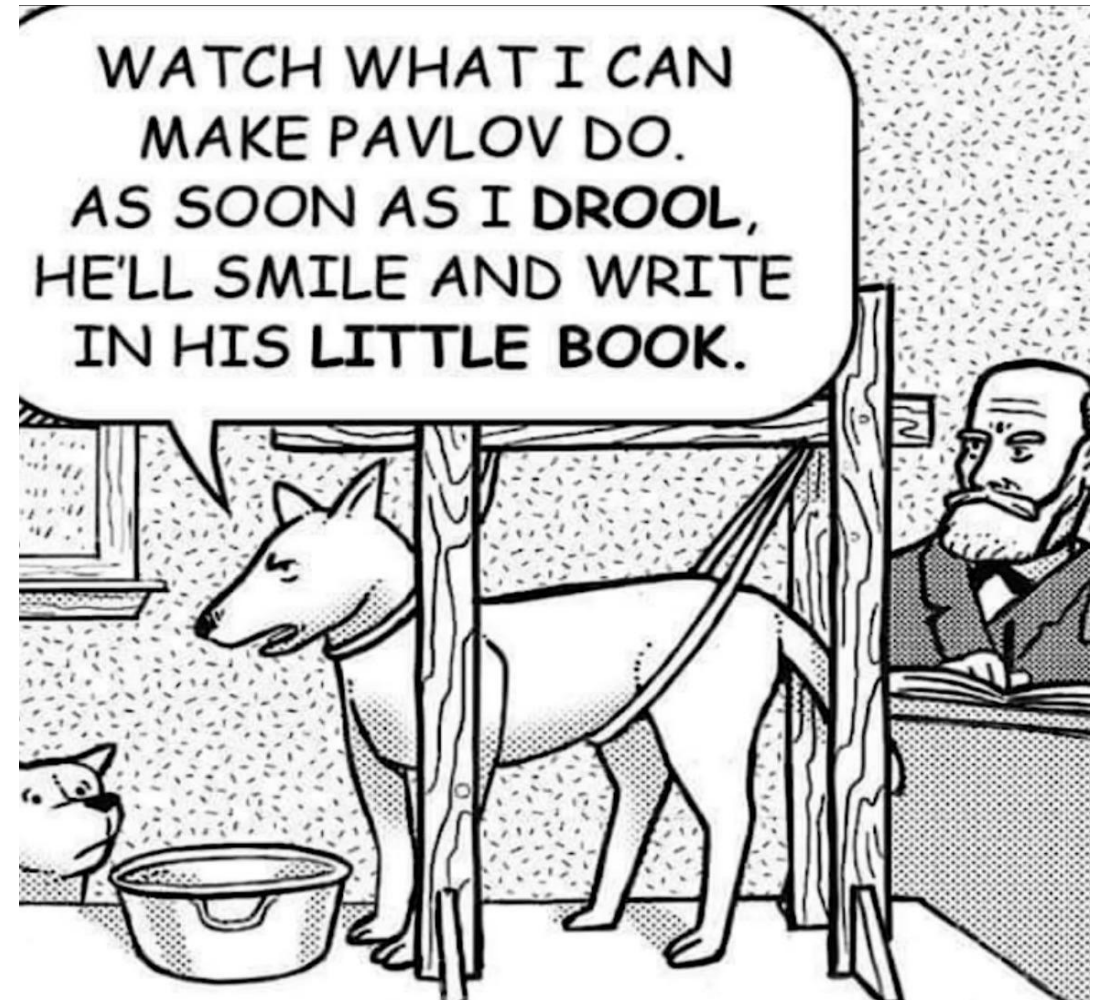
**SALIVATION
(CR)**







THEJENKINSComic



Everyday occurrences using Classical Conditioning



If you find yourself cringing at the mere sight of a hypodermic needle, your cringing is a CR to the CS of the needle. The pain of the shot would be the original UCS.

- Does your mouth water when you merely *see* an advertisement for your favorite food on television?
- Do you feel anxious every time you hear the high-pitched whine of the dentist's drill?
- The familiar ping of a notification can make your heart pump with excitement, or it could induce stress.
- **Neutral Stimulus:** Notification Sound
- **Unconditioned Response:** No emotional reaction
- **Natural Stimulus:** Something that triggers an emotional response
- **Natural Response:** Emotional Response (stress or excitement)
- **Conditioned Stimulus:** Notification Sound
- **Conditioned Response:** Emotional Response (stress or excitement)

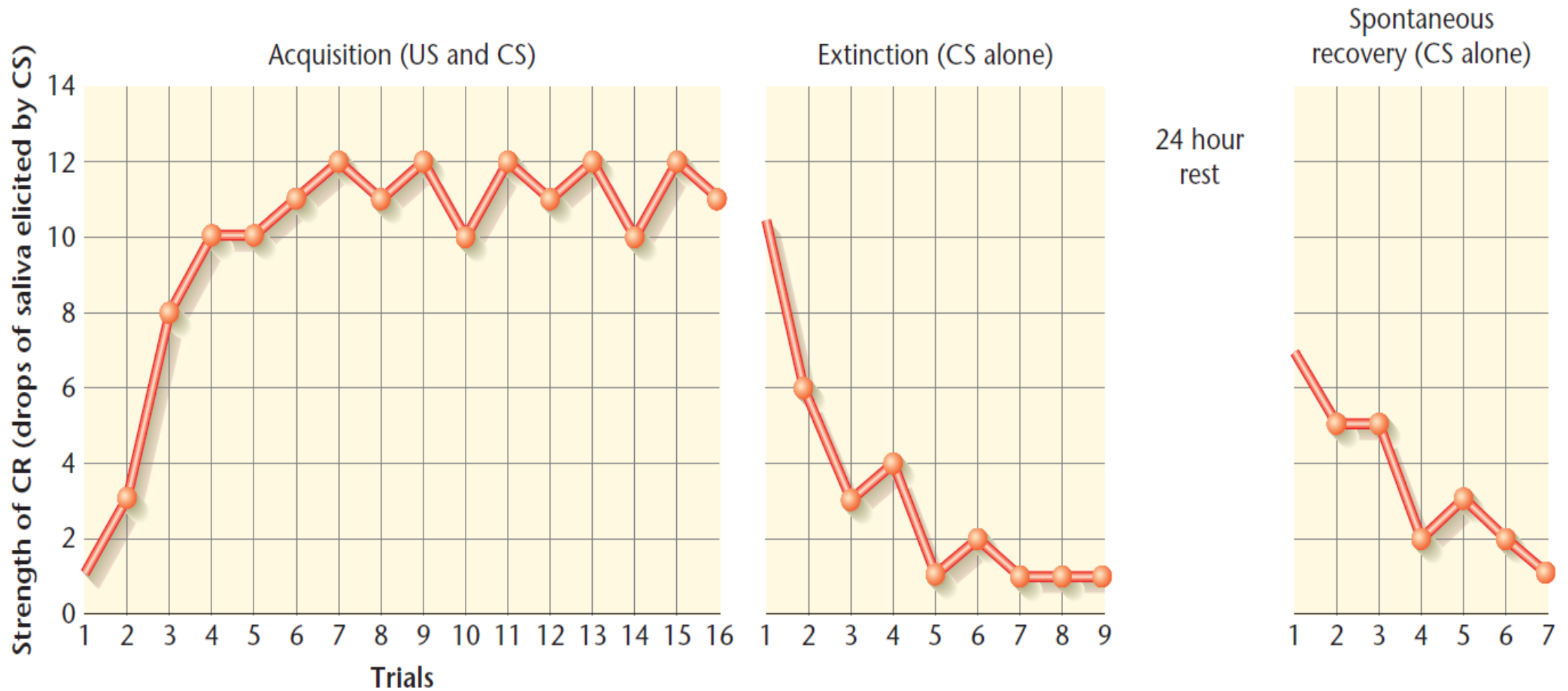
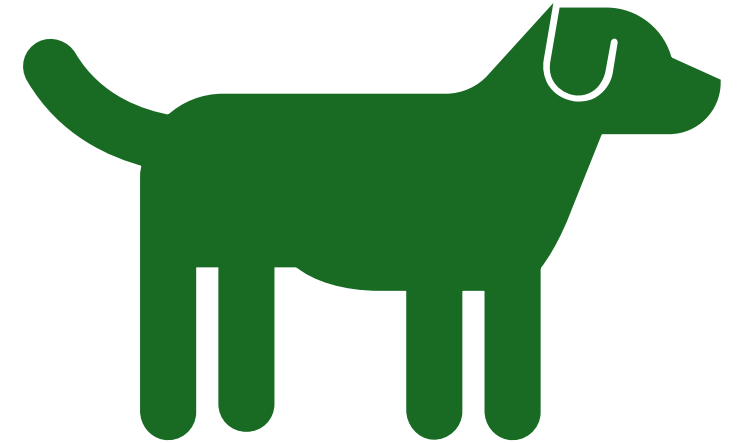


Figure 7.2 Acquisition and Extinction of a Conditioned Response. *The curve in the panel on the left depicts the acquisition phase of an experiment. Drops of saliva in response to the CS (before the onset of the US) are plotted on the vertical axis; the number of trials is plotted on the horizontal axis. After 16 acquisition trials, the experimenter switched to extinction; the results are presented in the panel in the middle. The panel on the right shows spontaneous recovery of the response after a 24 hour rest period. (Adapted from Conditioned Reflexes, by E. P. Pavlov. Copyright © 1927 by Oxford University Press. Reprinted by permission of Oxford University Press.)*

Few basic principles about the process

1. The CS must come *before* the UCS. If Pavlov sounded the metronome just after he gave the dogs the food, they did not become conditioned (Rescorla, 1988).
2. The CS and UCS must come very close together in time — ideally, no more than 5 seconds apart. When Pavlov tried to stretch the time between the potential CS and the UCS to several minutes, no association or link between the two was made. Too much could happen in the longer interval of time to interfere with conditioning (Pavlov, 1926; Ward et al., 2012; Wasserman & Miller, 1997).
3. The neutral stimulus must be paired with the UCS several times, often many times, before conditioning can take place (Pavlov, 1926).
4. The CS is usually some stimulus that is distinctive* or stands out from other competing stimuli. The metronome, for example, was a sound that was not normally present in the laboratory and, therefore, distinct (Pavlov, 1927; Rescorla, 1988).



- **Stimulus generalization**
- dogs that had been trained to have a conditioned response to a certain tone, would show the same response to a tone that was slightly higher or lower in pitch.
- **Response generalization:** the more similar the new stimuli are to the original CS, the more likely they are to evoke the same response.
- Accounts in part for a human or animal's ability to react to novel stimuli that are similar to familiar ones – an ability that is clearly adaptive
- **Stimulus Discrimination and Differential conditioning:** another possible consequence of classical conditioning is a decrease in the probability or magnitude of a behavioral response – this is inhibitory conditioning.

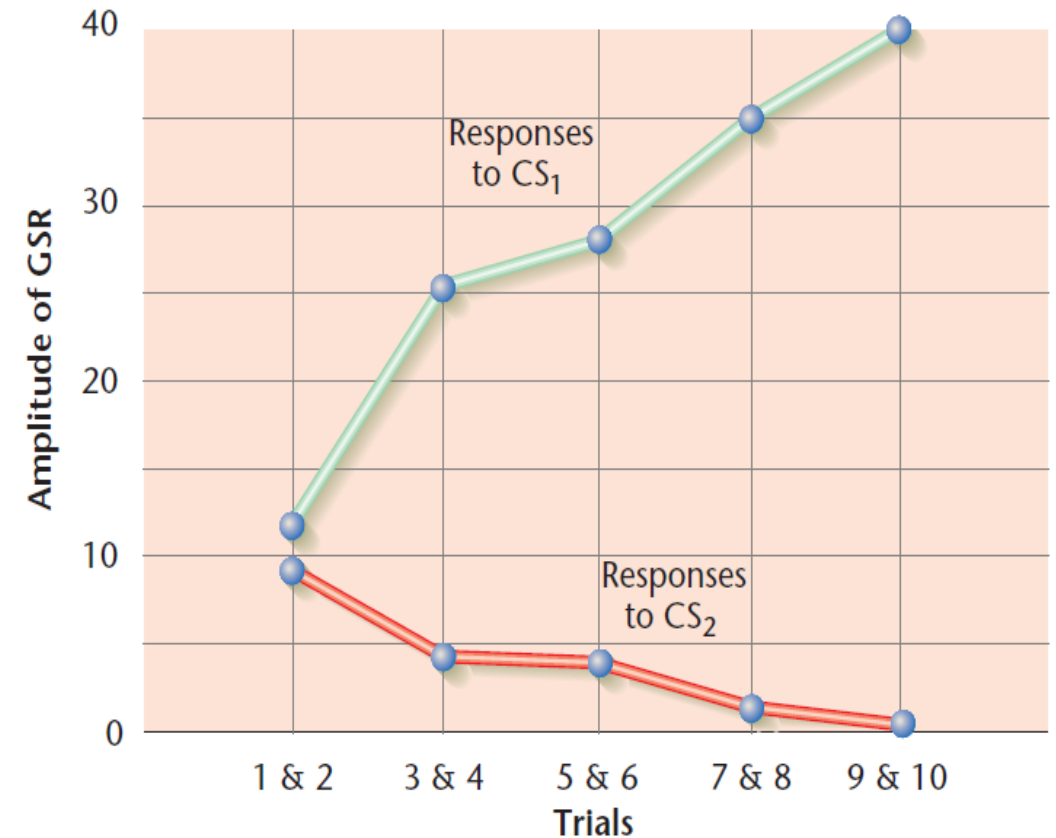


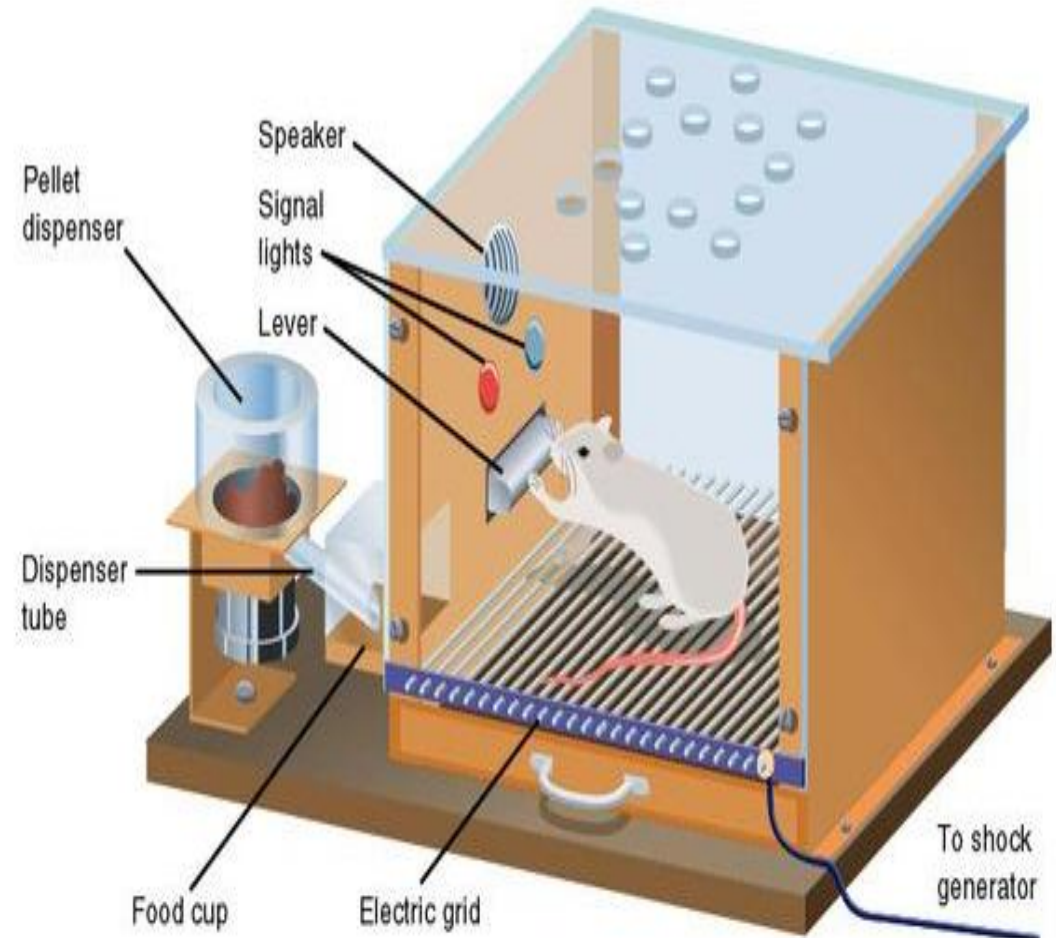
Figure 7.4 Conditioned Discrimination. *The discriminative stimuli were two tones of clearly different pitch (CS₁ = 700 Hertz and CS₂ = 3,500 Hertz). The unconditioned stimulus, an electric shock applied to the left forefinger, occurred only on trials when CS₁ was presented. The strength of the conditioned response, in this case the GSR, gradually increased following CS₁ and extinguished following CS₂. (Adapted from "Differential Classical Conditioning: Verbalization of Stimulus Contingencies," by M. J. Fuhrer & P. E. Baer, reprinted by permission from Science, Vol. 150, December 10.*

Biological constraints

- Early behaviorists assumed that the laws of learning were the same for all species. Moreover, they assumed that any CS could be associated with any US through classical conditioning.
- Early behaviourists firmly on the nurture side of the nature–nurture debate: what an organism learns, depends entirely on its experiences with the environment. – Babbling behaviour (Watson, 1913).
- Konrad Lorentz, Nikolaas Tinbergen, and Karl von Frisch made discoveries that revealed powerful biological predispositions in human and animal behavior (Tinbergen, 1951)

Skinner's Operant Chamber

- When a food-deprived rat is placed in the box, its behavior at first is spontaneous and random. The rat is active, sniffing, poking, and exploring its environment. These behaviors are emitted, not elicited; in other words, the rat is not responding to any specific stimulus in its environment.
- At some time during this activity, the rat will depress a lever or bar located on one wall of the Skinner box, causing a food pellet to drop into a trough.
- The food is a reinforcer for the behavior of depressing the bar.



OPERANT CONDITIONING

- **Or Instrumental conditioning:** instrumental – as learning some action (or behaviour) is instrumental in bringing about a change in the environment that makes the action (or behaviour) more or less likely to occur again in future
- E.g. getting a reward
- **Reinforcer:** An environmental event that is the consequence of an instrumental response and that makes that response more likely to occur again- **increases the likelihood that behaviour will occur again**
- **Positive reinforcer:** Stimulus or event which when it follows a response increases the likelihood that the response will be made again – bar pressing is followed by food
- the reinforcer has to be contingent upon the instrumental response

Operant Conditioning



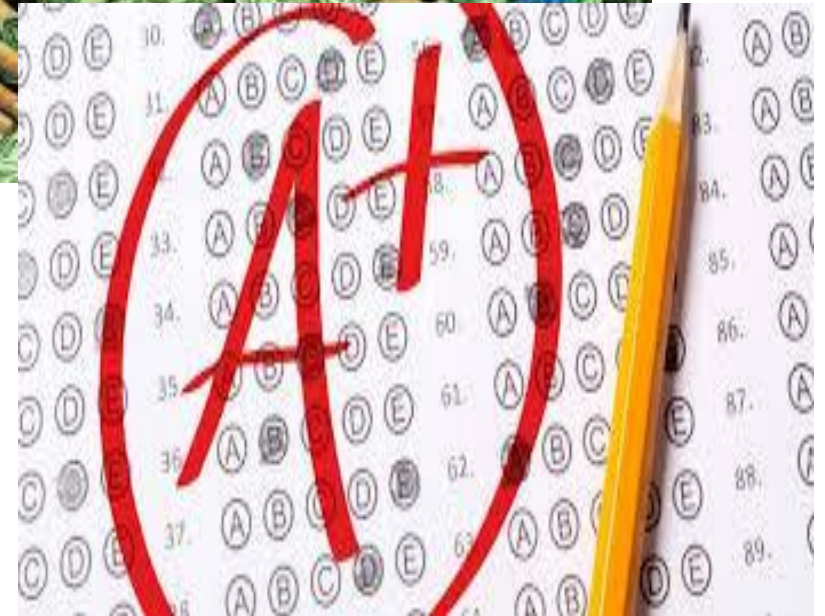
Positive reinforcement
introduced to increase
a behavior



Negative reinforcement
introduced to decrease
a behavior

Reinforcers

- The response results in the occurrence of the reinforcer – that is responses that pay off – are repeated
- E.g. food, water, praise, prize – when contingent on behaviour
- Primary reinforcers are stimuli that are naturally reinforcing because they are not learned and directly satisfy a need, such as food or water.
- Secondary reinforcers are stimuli that are reinforced through their association with a primary reinforcer, such as money, school grades. They do not directly satisfy an innate need but may be the means. So a secondary reinforcer can be just as powerful a motivator as a primary reinforcer.



Negative reinforcer

- A stimulus or event which, when its cessation or termination is contingent on a response, increases the likelihood that the response you want them to learn will occur again
- The word negative (-ve) refers to the fact that the response causes the **termination of the event**
- E.g. bar pressing terminates or prevents shock, bawling by the boss will stop on doing good work

- **Response tend to become more likely**

Punishment

- A stimulus or event whose onset is contingent on a response and thereby decreases the likelihood that the response will occur again
- E.g. the boss shouting if the job is not done well, Traffic fines, demerits, beatings – when contingent on behaviour ending,
- **Responses followed by punishment do not tend to be repeated**



Successive Approximation: The Shaping of Behavior

- Animals and humans demonstrate many more complex operant behaviors that have a much lower probability of occurrence in the normal course of events.
- How are these complex behaviors learned?
- How can an experimenter or a parent reinforce and condition a pigeon or a child to perform behaviors that are not likely to occur spontaneously?



1. The pigeon was reinforced with food when it merely turned toward the designated spot.
2. Then reinforcement was withheld until the pigeon made some movement, however slight, toward the spot.
3. Reinforcement was given only for movements that brought the pigeon closer to the spot.
4. After that, the pigeon was reinforced only when it thrust its head toward the spot. Finally, the pigeon was reinforced only when its beak touched the spot.

Although this sounds like a time-consuming process, Skinner conditioned pigeons in fewer than 3 minutes.

- Skinner's explanation of language acquisition in children.

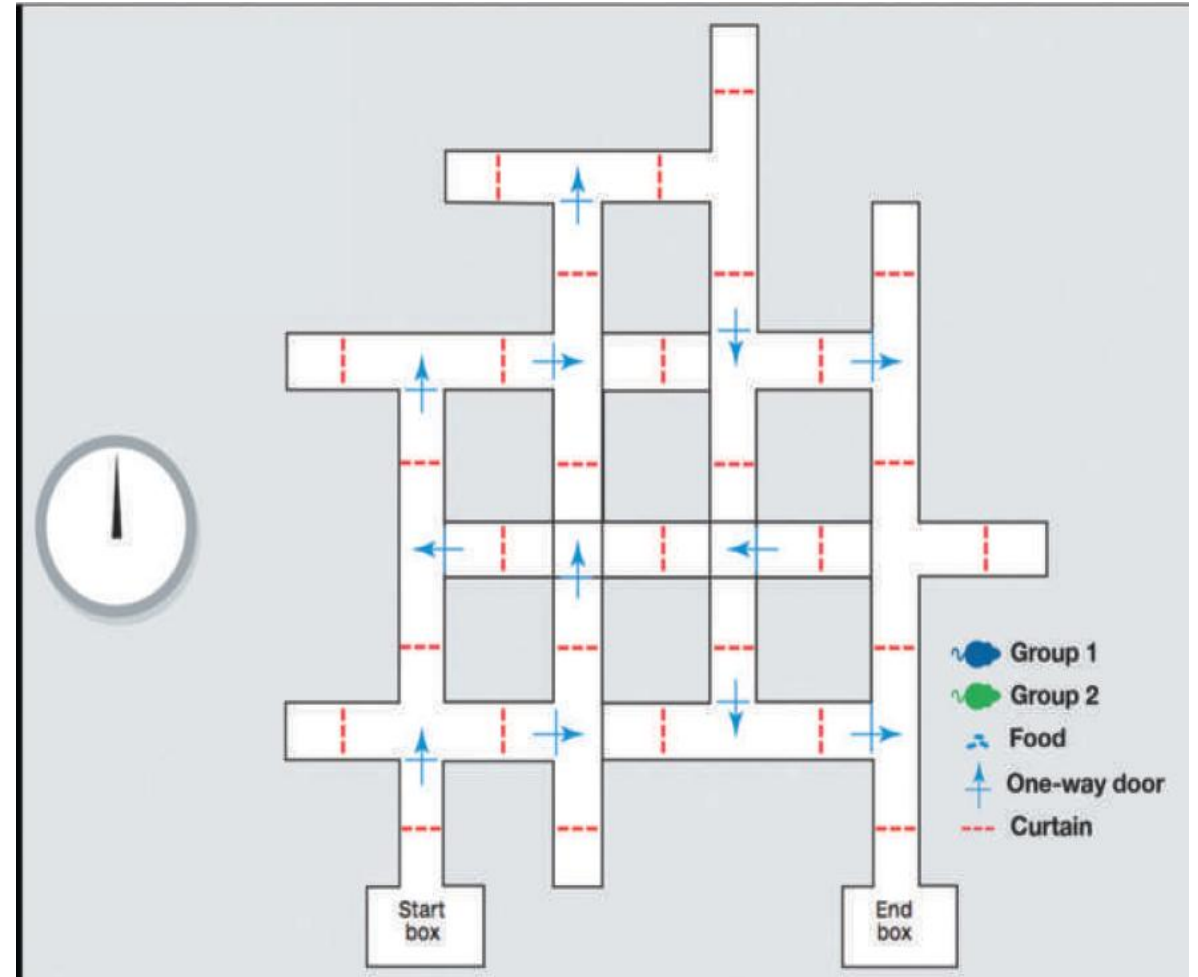
New use: Pigeon collecting waste

- Skinner believed that most human and animal behavior is learned through operant conditioning.
- E.g. learning among babies - An infant initially displays random, spontaneous behaviors, only some of which are reinforced (rewarded with food or hugs or toys, for example) by parents, siblings, or caregivers. As the infant grows, the positively reinforced behaviors, those of which the parents approve, will persist, whereas those of which the parents disapprove will be extinguished or discontinued. The concept is the same with the rat in the Skinner box.

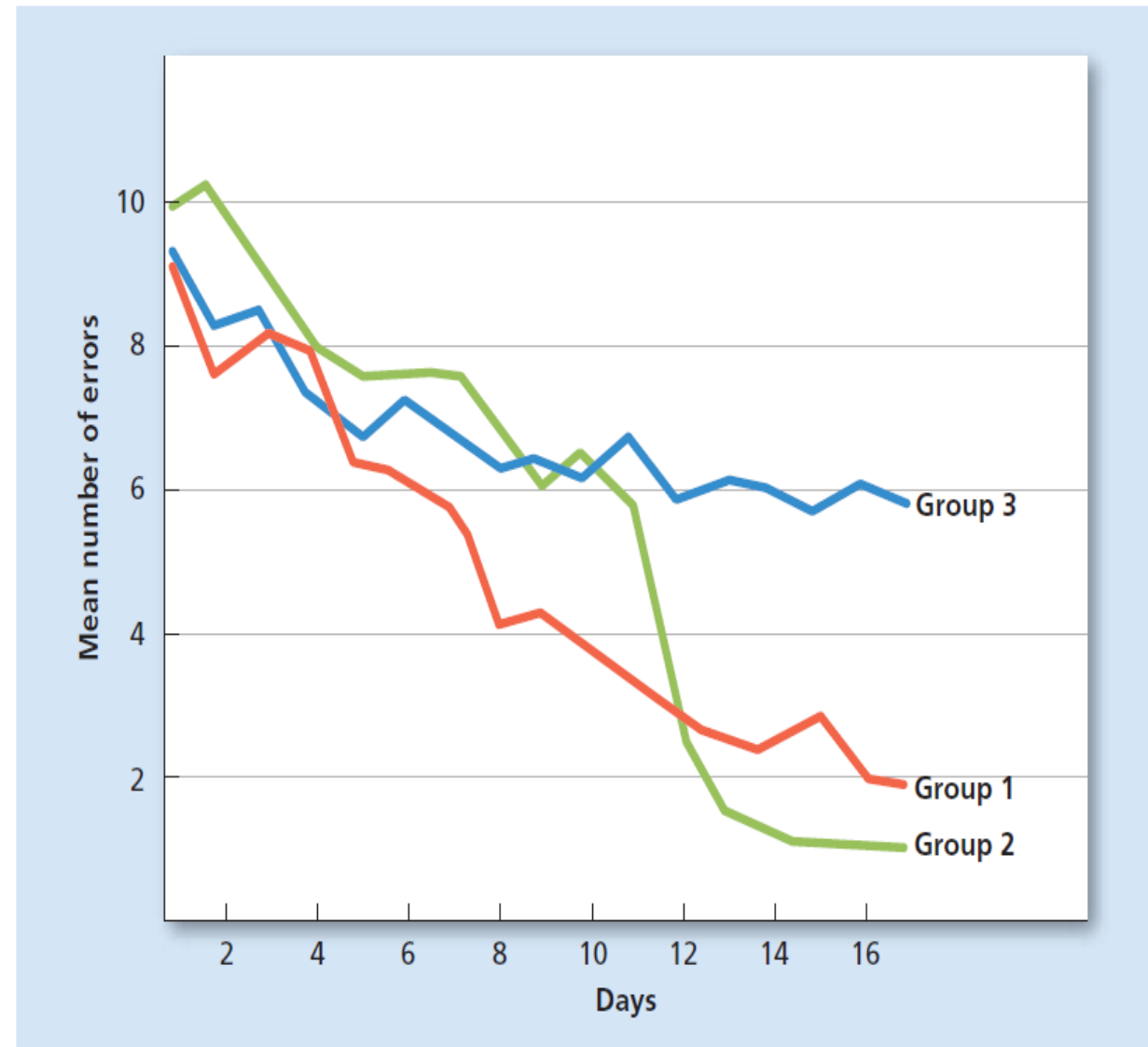


Cognitive Learning Theory

- Tolman's Maze-Running Rats : Latent Learning
- Teaching three groups of rats the same maze, one at a time (Tolman & Honzik, 1930).
- Group 1 : each rat was placed in the maze and reinforced with food for making its way out the other side. The rat was then placed back in the maze, reinforced upon completing the maze again, and so on until the rat could successfully solve the maze with no errors
- Group 2: The second group of rats never received any reinforcement upon exiting the maze. Put back in again and again, until the 10th day of the experiment- reinforcement given
- Group 3: control group, was also not reinforced and was not given reinforcement for the entire duration of the experiment



- The second group of rats, *should* have then taken as long as the first group to solve the maze. Instead, they solved the maze almost immediately
- The rats had learned where all the blind alleys, wrong turns, and correct paths were and stored this knowledge away as a kind of “mental map,” or *cognitive map* of the physical layout of the maze.
- **Cognitive map** – a mental understanding of an environment, formed through trial and error as well as observation. The concept is based on the assumption that an individual seeks and collects contextual clues, such as environmental relationships, rather than acting as a passive receptor of information needed to achieve a goal. (APA, 2018)
- Rats in the second group had learned and stored that learning away mentally but had not *demonstrated* this learning because there was no reason to do so.
- **Latent learning:** learning that is acquired without conscious effort, awareness, intention, or reinforcement and is not manifested as a change in performance until a specific need for it arises. (APA, 2018)



Kholer's Insight Theory

- (Köhler, 1925), he set up a problem for one of the chimpanzees.
- Sultan the chimp was faced with the problem of how to get to a banana that was placed just out of his reach outside his cage.

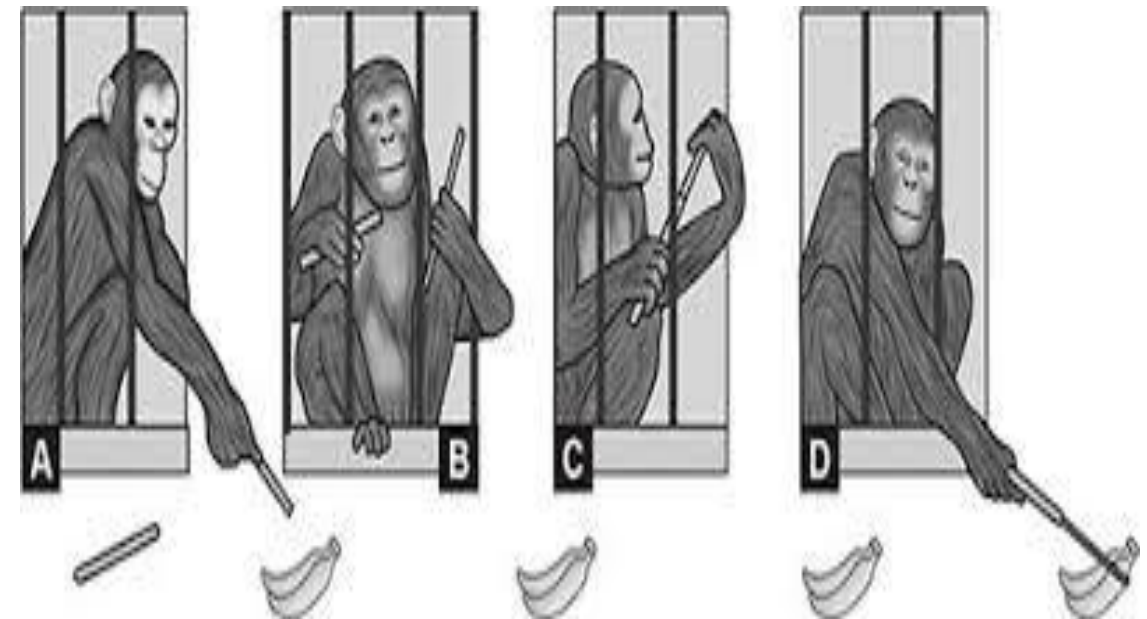


Köhler's Insight Theory

- Köhler called Sultan's rapid "perception of relationships" **insight** and determined that insight could not be gained through trial-and-error learning alone (Köhler, 1925).

Insight requires a sudden

- "coming together" of all the elements
- of a problem in a kind of "aha" moment that is not predicted by traditional animal learning studies.
- 149162536496481100...
- Perceptual reorganization of elements in the environment such that new relationships between objects are suddenly seen



Observational Learning: “monkey see, monkey do.”

- Learning of new behavior through watching the actions of a model (someone else who is doing that behavior) – form of latent learning.
- **Imitation behaviour:** a response that is like the stimulus that triggered it
- Sometimes that behavior is desirable, and sometimes it is not
- **Bandura’s Bobo Doll experiment (1961)**



Ah, but would that child have imitated the model if the model had been punished? Wouldn't the consequences of the model's behavior make a difference?

- The tendency for some movies and television programs to make “heroes” out of violent, aggressive “bad guys” is particularly disturbing in light of these findings. In fact, Bandura began this research to investigate possible links between children’s exposure to violence on television and aggressive behavior toward others.



- Prosocial behavior, which is behavior aimed at helping others, has also been shown to be influenced by media consumption. Studies have shown that when children watch media that models helping behavior, aggressive behavior decreases and prosocial behavior increases (Anderson et al., 2015; Prot et al., 2014).



cognitive abilities necessary for observational learning to occur (Bandura, 1977, 2001).

- The learner must be able to
 - (1) pay attention to the model's behavior and observe its consequences,
 - (2) remember what was observed,
 - (3) be able to reproduce the behavior, and
 - (4) be motivated to do so.
- In other words: observational learning involves the ability to imagine and anticipate – thoughts and intentions are essential.
- Bandura's 'agentic perspective' draws our attention to the fact that cognitions motivate actions, and that a sense of **self-efficacy** (an individual's belief in their own effectiveness) is essential for complex and social learning.
- If you believe that you are simply incapable of giving a good presentation in class, you are unlikely to motivate



Raccoons commonly dunk their food in and out of water before eating. This “washing” behavior is controlled by instinct and difficult to change even using operant techniques.

Biological Constrains

- As the coins became more and more associated with food, the animals began to drift back into the instinctual patterns of behavior that they used with real food.
- Raccoons rub their food between their paws and dip it in and out of water.
- Pigs root and throw their food around before eating it.
- The Brelands called this tendency to revert to genetically controlled patterns **instinctive drift**.

Schedules of Reinforcement

- Skinner pointed out that in everyday life outside the psychology laboratory, our behavior is rarely reinforced every time it occurs (**continuous reinforcement**). E.g. babies, batsmen, bagger, singers, etc.
- 1. **Fixed-interval schedule:** interval might be 1 minute, 3 minutes, or any other fixed period of time. E.g. scheduled tests, monthly salary.
- 2. **Fixed-ratio schedule:** reinforcers are given only after the organism has made a specified number of responses. For example, the experimenter could reinforce after every 10th or 20th response. E.g. sales or produce jobs
- faster rate of responding than does the fixed-interval schedule.
- Nearly all reinforcement begins with continuous reinforcement then move towards intermittent.

Superstitious behaviour

- But everyday life doesn't always permit a fixed-interval or fixed-ratio reinforcement schedule.
- 3. **Variable-interval schedule:** the reinforcer might appear after 2 hours in the first instance, after 1 and a 1/2 hours the next time, and after 2 hours and 15 minutes the third time. E.g. fishing experiences
- 4. **Variable-ratio schedule:** reinforced after every n^{th} response on the average.
3 – 6 – 10 , so on.
- After a high mean is reached, say 500, responses become extremely resistant to extinction.
- Extinction: the tendency of a previously acquired response to become progressively weakened upon non-reinforcement.
e.g. playing with slot machines.

What are some
possible waste
disposal
learning
techniques?

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- drift back into the instinctual patterns of behavior that they used with real food.

Raccoons

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